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Spurious Steady State from dt-Dimensional Ambiguity in Modular ODE Coupling: A Failure Mode, Detection Protocol, and Cardiovascular Demonstration

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Spurious Steady State from dt-Dimensional Ambiguity in Modular ODE Coupling: A Failure Mode, Detection Protocol, and Cardiovascular Demonstration

Abstract

Multi-organ physiological simulators commonly couple organ modules through threshold-gated discrete events that modify shared state variables. We characterize spurious steady state — a failure mode in which such events carry a dt-dimensional ambiguity (increment per step vs rate per second), producing a stable but incorrect steady state that standard convergence diagnostics cannot detect. Using an 11-organ canine cardiovascular simulation as a demonstrative testbed, we trace one instance: a heart rate modifier implemented as a fixed increment per time step (bpm/step) rather than as a time-normalized rate (bpm/s). At fine integration steps ($dt = 0.001$ s), this produced an effective HR injection rate of 10,000 bpm/s — 1,000 times the intended magnitude. The resulting steady-state MAP bias reached +44.7 mmHg before heart rate saturated at its physiological ceiling of 180 bpm. Standard convergence diagnostics (dt refinement, steady-state detection, parameter sweeps) detected the anomaly but misidentified it as physiological saturation. Correction required three changes across production files (7 lines total): scaling discrete deltas by dt on emission, eliminating a redundant parallel path, and applying exponential rate conversion to the SVR-multiply channel. Post-fix MAP variance across a dt sweep (0.1–0.001 s) fell from 44.7 mmHg to ≤ 2.21 mmHg. A minimal 2-variable toy model — containing no physiological parameters — reproduces

the entire bias pattern, suggesting the failure mode is intrinsic to the coupling design rather than contingent on cardiovascular physiology. The findings alert developers of threshold-gated discrete event architectures to a latent dimensional inconsistency that routine numerical verification may miss.

1. Introduction

Modular simulation architectures — in which autonomous subsystem modules communicate through shared state variables — are a dominant design pattern across domains from physiological modeling [1] to multi-physics co-simulation [2]. In such architectures, a common inter-module communication mechanism is the threshold-gated discrete event: Module A monitors a state variable, and when a threshold is crossed, emits an event that modifies a state variable in Module B. This pattern appears in physiological engines (e.g., baroreflex-mediated heart rate adjustment), robotic control systems (e.g., sensor-triggered actuator commands), and discrete-event simulation frameworks (e.g., DEVS-based coupling [3]). When the receiving module integrates its dynamics with explicit Euler, the event's dimensional interpretation — whether it represents a rate (per-second) or a fixed increment (per-step) — becomes critical.

Several groups have documented the numerical challenges of closed-loop cardiovascular simulation. van Osta et al. [4] demonstrated that closed-loop regulation substantially

extends equilibration time in zero-dimensional cardiovascular models compared to open-loop configurations. Tłalka et al. [5] identified stability concerns with explicit integration in baroreflex-coupled models, motivating the use of implicit or semi-implicit schemes. Ursino [6], in his foundational work on carotid baroreflex modeling, employed implicit Gear-style solvers as a matter of course.

Yet the specific failure mode of threshold-gated discrete inter-module events — a common design pattern in many simulation engines — has received little attention. In this pattern, Module A monitors a state variable, and when a threshold is exceeded, emits a discrete event that adds or multiplies a state variable in Module B. The question addressed here is simple: can a seemingly innocuous dimensional oversight in such discrete events produce a stable but physiologically wrong steady state that standard convergence checks do not detect?

We characterize spurious steady state: a failure mode in which modular ODE systems with threshold-gated discrete events converge to a stable but incorrect steady state while passing all standard convergence diagnostics. Using Virtual Vet — an 11-organ canine cardiovascular simulation — as a demonstrative testbed, we trace one instance through its discovery, root cause analysis, dimensional diagnosis, and four-condition isolation. We then generalize by (i) confirming the mechanism is intrinsic to the coupling pattern

through a domain-independent toy model (Section 3.5), (ii) extracting a three-level detection protocol applicable to any modular simulation (Section 4.2), and (iii) proposing temporal type metadata for discrete-event interfaces — a dimension absent from existing coupling standards (Section 4.3).

2. Methods

2.1 Virtual Vet Platform

Virtual Vet is a closed-loop, 11-organ canine cardiovascular simulation implemented in Python 3, designed for veterinary clinical education. Organ modules — heart, lung, kidney, gut, liver, endocrine, neural (baroreflex), immune, coagulation, lymphatic, and fluid — share a blood compartment and communicate through a unified dispatch interface. All parameter modifications are mediated through a single function that resolves dot-path targets (e.g., `heart.heart_rate`), accepting operations of type multiply, add, or set:

```
@dataclass(frozen=True)
class FactorCommand:
    target: str    # dot-path like "heart.heart_rate"
    op: Literal["multiply", "add", "set"]
    value: float
```

This design ensures a uniform write interface across all physiological subsystems — baroreflex, diseases, drugs, and treatments all route through the same dispatch. The

sequential step order is hard-coded: heart $\rightarrow$ lung $\rightarrow$ kidney $\rightarrow$ gut $\rightarrow$ liver $\rightarrow$ endocrine $\rightarrow$ neuro $\rightarrow$ immune $\rightarrow$ coagulation $\rightarrow$ lymphatic $\rightarrow$ fluid, executing each module's `compute(dt, ...)` method once per time step using forward Euler integration [7].

Default integration parameters: $dt = 0.01$ s, body weight 20 kg, age 1095 days (adult canine). At baseline ($DC = 25$, no disease), the post-fix simulation produces $MAP = 100.9$ mmHg and $HR = 108.5$ bpm — within the normal canine resting range of $MAP = 85\text{--}120$ mmHg and $HR = 60\text{--}140$ bpm reported by VetFolio and Acierno et al. [8, 9].

2.2 Dual Baroreflex Architecture

Two parallel mechanisms mediate baroreflex heart rate control, both targeting `heart.heart_rate`:

Continuous path (`heart.py: _baroreceptor_feedback`). The feedback controller normalizes MAP deviation by the setpoint ($error = (MAP_target - MAP) / MAP_target$) and drives two antagonistic first-order channels. Time constants are set to $\tau_{symp} = 2$ s and $\tau_{para} = 5$ s, chosen to match canine sympathetic and parasympathetic response latencies. The HR delta is:

```
HR_para = -P_para * 15.0 * max(0, -error)
HR_symp = P_symp * 50.0 * max(0, error)
chemo_HR = chemo_drive * 15.0
```

```
HR_delta = (HR_para + HR_symp + chemo_HR) × dt
```

This path is dimensionally correct: HR_para and HR_symp are rates in bpm/s, multiplied by dt to yield the per-step increment.

Discrete path (neuro.py: FactorCommand emission). The neuro module monitors blood gases, pain, seizure, and consciousness. When the chemoreceptor drive exceeds a threshold ($\text{chemo_drive} > 0.01$), it emits FactorCommands:

```
net_HR_add = pain_HR_add + seizure_HR_add + cns_HR_add + chemo_HR_add
if abs(net_HR_add) > 0.1:
    factor_commands.append(
        FactorCommand("heart.heart_rate", "add", net_HR_add)
    )
```

Where $\text{chemo_HR_add} = \text{chemoreceptor_drive} \times 10.0$.

The critical distinction: the continuous path interprets its output as a rate (bpm/s), while the discrete path (as originally implemented) interpreted net_HR_add as a fixed increment per step (bpm/step). Under normal conditions (no pain, seizure, or CNS failure), the dominant driver was chemo_HR_add.

2.3 Convergence Study Design

We performed a systematic dt sweep across dt in {0.1, 0.05, 0.02, 0.01, 0.005, 0.002, 0.001} s with a fixed 60 s simulation window. To evaluate chemoreflex engagement

across a range of physiological conditions, we varied the pulmonary diffusion coefficient (DC) from 5 to 25 — lower values produce more severe hypoxemia. All simulations began from identical initial conditions with a 30 s equilibration period before data collection.

2.4 Experimental Data Sources

Three experiment generations are reported: - Exp6: Original (buggy) code — FC deltas unscaled by dt - Exp7: FC dt-scaling fix only — deltas multiplied by dt on emission - Exp8: Complete fix — FC dt-scaling + chemoreflex moved to continuous path

Post-fix clinical validity was assessed against Tucker et al. [10], which reported HR and MAP responses to graded hypoxia in conscious dogs.

2.5 Toy Model: Domain-Independent Minimal Demonstration

To confirm that the spurious steady state arises from the coupling pattern itself — not from cardiovascular physiology — we constructed a minimal 2-variable ODE system:

Plant:	$dx/dt = -k \cdot (x - x_0)$	[continuous homeostasis]
Controller:	when sensor > threshold $\rightarrow$ FC("x", "add", K)	
	BUGGY: K [unit/step]	FIXED: K·dt [unit/s]
Saturation:	$x \leq x_{\text{ceiling}}$	[truncation ceiling]

The plant represents any continuous system maintaining a setpoint x_0 with relaxation rate

k. The controller mimics threshold-gated discrete events: when an external signal exceeds threshold, it emits a fixed-magnitude event modifying x . No physiological assumptions are embedded — the parameters ($k = 0.25$, $K = 0.5$, $x_0 = 100$, $x_{\text{ceiling}} = 180$) were chosen purely to produce the same qualitative bias pattern as Virtual Vet for visual comparison.

3. Results

3.1 Discovery: Anomalous dt-Dependent Bias

Under the original code, a simple dt-sweep convergence study [11] (DC=10, mild hypoxia) revealed a striking pattern: steady-state MAP increased systematically as dt decreased, reaching +44.7 mmHg above nominal by dt=0.02 s (Table 1).

Table 1: Pre-fix MAP and HR across dt sweep (DC=10)

dt (s)	MAP (mmHg)	MAP bias (mmHg)	HR (bpm)	Regime
0.1	111.0	+11.0	108.5	Unsaturated
0.05	121.1	+21.1	129.8	Unsaturated
0.02	144.7	+44.7	180.2	Saturated
0.01	144.7	+44.7	180.0	Saturated
0.005	144.7	+44.7	180.0	Saturated
0.001	144.7	+44.7	180.0	Saturated

The pattern is striking for two reasons. First, the bias increases as dt decreases — the opposite of what is expected from truncation error, which scales as $O(dt)$ for explicit

Euler [7]. Second, below $dt \approx 0.02$ s, the system plateaus at a precisely constant value. The product $\text{bias} \times dt$ in the unsaturated regime is nearly constant (1.14 ± 0.04), implying $\text{bias} \sim 1/dt$. Figure 2 shows this MAP bias vs dt relationship across the full dt sweep on both log-log and semilog scales, revealing the unsaturated drift regime and the saturated plateau.

The strict constancy of MAP in the saturated regime (144.7 mmHg across $dt = 0.02$ to 0.001 s) reflects the compounding effect of the HR ceiling on cardiac output: when HR = 180 bpm (the physiological ceiling), $CO = HR \times SV$ is similarly capped, and $MAP = CO \times SVR$ then inherits this saturation through the circulation. Once both HR and CO saturate, further dt reduction cannot increase MAP further.

This behavior superficially resembles the fixed-point destabilization described in sequential coupled analyses [12], and we initially pursued this explanation — noting, however, that Kim's framework addresses operator splitting in coupled PDE systems (poromechanics), not discrete inter-module events in modular ODE engines, so the analogy is heuristic rather than formally applicable. The parameter insensitivity of the bias — identical across baroreflex gains from $0.5\times$ to $8.0\times$, MAP initializations, and body masses — suggested a mechanism more fundamental than gain-mediated instability.

3.2 Diagnosis: Exclusion of State Pollution

A critical clue came from swap experiments comparing heart→neuro and neuro→heart orderings. In same-process comparisons, the two orderings sometimes showed large differences (up to 44.7 mmHg), prompting speculation that module ordering determined accuracy. However, subprocess isolation — executing each ordering in a fresh Python interpreter — showed they were identical under steady-state baseline conditions:

dt (s)	heart→neuro MAP (mmHg)	neuro→heart MAP (mmHg)	Δ
0.01	144.742	144.776	0.034

The Δ of 0.034 mmHg is floating-point noise. The apparent ordering-dependence in same-process comparisons was an artifact of Python class-level state carryover between runs. This ruled out sequential coupling as the primary mechanism under steady state — both orderings converged to the same erroneous steady state.

However, this conclusion was incomplete. Under a hemorrhage challenge (400 mL blood loss over 120 s), subprocess-isolated ordering comparison revealed a different picture: the maximum Δ reached -2.84 mmHg during the transient phase ($t = 58$ s), despite the two orderings converging to near-identical values at steady state ($\Delta_{\text{final}} = +0.028$ mmHg). The ordering difference was real but transient — it appeared during active compensation and vanished once both systems reached saturation at HR = 180 bpm.

This has a direct implication for the paper's core claims: the FC dt-scaling fix eliminates the bias under steady-state conditions, but hemorrhage-induced transients reveal residual ordering sensitivity. The primary claim "ordering is irrelevant to the bias" must be qualified as "ordering is irrelevant under steady-state baseline conditions." The ordering sensitivity in hemorrhage transient was present in both the original buggy code and the fixed code — it is a genuine property of the sequential Euler architecture, not the dt-dependent bias, and is outside the scope of the present fix.

3.3 Diagnosis: Dimensional Analysis of FactorCommand Emission

With ordering ruled out, we traced the mechanism to the FactorCommand emission in `src/neuro.py`. The key observation was that `net_HR_add` was emitted as a fixed increment:

```
# Original (buggy): fixed increment per time step
factor_commands.append(FactorCommand("heart.heart_rate", "add", net_HR_add))
```

The value `net_HR_add` had units of bpm/step — but this was not apparent from the code, since the variable was computed from dimensionless products like `chemoreceptor_drive` $\times$ 10.0. The continuous baroreflex path in `heart.py` [6], by contrast, explicitly multiplied by `dt`.

The consequence is straightforward. At each step, an increment of `K` bpm is injected into

heart_rate. In T seconds of simulation, the total injected HR is:

$$\text{Total HR injection} = K \times N_{\text{steps}} = K \times T/\text{dt} \sim 1/\text{dt}$$

With $\text{chemo_drive} \approx 0.01$ near threshold: $\text{net_HR_add} = 0.01 \times 10.0 = 0.1$ bpm/step. At $\text{dt} = 0.01$ s (default), this produces an effective injection rate of 10 bpm/s — comparable to the continuous path's rate. But at $\text{dt} = 0.001$ s, the same threshold-driven value produces 100 bpm/s. In the limit $\text{dt} \rightarrow 0$, the injection rate diverges to infinity.

This is not a numerical stability issue in the ODE sense — it is a dimensional analysis error: a per-step quantity was not normalized by dt before being treated as a rate.

A second finding emerged during code review: the chemoreflex HR effect was being applied through both the continuous baroreflex path (heart.py) and the discrete FC path (neuro.py), creating a redundant double-counting:

```
# net_HR_add in neuro.py included chemo_HR_add...
net_HR_add = (pain_HR_add + seizure_HR_add + cns_HR_add + chemo_HR_add)
# ...while heart.py _baroreceptor_feedback also handled chemoreflex
chemo_HR = chemoreceptor_drive * 15.0 # bpm/s
HR_delta = (HR_para + HR_symp + chemo_HR) * dt
```

The redundancy had masked the bug: when chemoreflex was withdrawn from either path alone, the other path continued to provide drive, making the effect of the dimensional error less obvious in isolated testing.

Table 2: Root cause summary

Factor	Description
Primary cause	FC HR delta was bpm/step, not bpm/s — missing dt normalization
Amplifier	Threshold gating: below-chemo_drive→0, above→fixed increment, making the dt-dependence discontinuous
Masking factor	Dual parallel paths (continuous + discrete) created redundancy that obscured the error
Detectability	Standard diagnostics (dt sweep, steady-state detection, parameter scan) all passed
Misleading clue	Same-process swap experiments showed false ordering dependence due to state pollution

3.4 Disentangling the Two Fixes

The 7-line fix comprises three independent corrections:

- A: multiply FC delta by dt on emission → dimensional normalization for HR-additive and RR-additive FC
- B: remove chemo_HR_add from net_HR_add and add continuous chemo path in heart.py
- C: apply exponential rate conversion to SVR-multiply FC ($SVR_{new} = SVR \times net_SVR_mult^{dt}$) to correct the analogous multiplicative dt-dependency in the SVR channel

Figure 3 shows the complete code diff for all three changes.

To determine their independent contributions, we conducted a four-condition isolation experiment using subprocess-isolated runs. Because the path consolidation (B) also changed the chemo gain constant (10 bpm/s on the FC path vs 15 bpm/s on the continuous path), a fourth condition W disambiguates the confound:

- X (baseline buggy): FC as bpm/step, chemo in net_HR_add, no continuous path - Y (A-only): $FC \times dt$, chemo in net_HR_add, no continuous path - W (A+path, gain=10): $FC \times dt$, chemo excluded, continuous path active (gain=10) - Z (A+B, current): $FC \times dt$, chemo excluded, continuous path active (gain=15)

Table 3: Four-condition isolation — MAP range across dt sweep (mmHg)

DC	X_range	Y_range	W_range	Z_range	A_contrib	B_contrib	C_contrib
25 (normal)	0.10	0.10	0.27	0.35	0.00	-0.17	-0.08
15 (mild)	2.52	2.52	2.70	2.80	0.00	-0.19	-0.09
10 (moderate)	31.99	0.41	0.37	0.51	**31.58**	0.03	-0.14
5 (severe)	20.84	2.20	2.20	2.21	**18.64**	-0.00	-0.01

$A_contrib = X_range - Y_range$; $B_contrib = Y_range - W_range$; $C_contrib = W_range - Z_range$. Figure 4 shows these comparisons as grouped bars for visual assessment across conditions and DC values.

Under mild conditions ($DC \geq 15$), all four conditions converge to essentially the same result because chemoreceptor drive is near zero — there is no excess HR drive to be injected, so the dt-dependent bias does not manifest regardless of the code path.

Under moderate and severe hypoxia ($DC = 10, 5$), the picture is clear: Operation A alone accounts for all of the improvement. Removing the FC dt-dependency (Y) reduces the MAP range from 31.99 mmHg to 0.41 mmHg at $DC=10$. The fourth condition W

resolves the gain confound: B_contrib (path consolidation alone, same gain constant) is 0.03 mmHg — confirming the architectural change has no effect on MAP range. C_contrib (gain change 10→15) is −0.14 mmHg, reflecting a slightly larger MAP range at the higher gain. Neither is material (< 0.2 mmHg).

Practical implication: The optimal configuration is A-only without B — FC dt-scaling alone fixes the bias with physiologically correct magnitudes. The continuous chemo path, while dimensionally correct and physiologically motivated, is not necessary for the bias elimination and slightly increases MAP range. However, it does improve physiological plausibility: at DC=5, HR rises to 96.9 bpm (Z) vs 93.3 bpm (Y), closer to the +8 bpm increase reported by Tucker et al. [10] for severe canine hypoxia.

The 7-line fix thus contains a design choice: operation A is the essential correction; operation B is a physiologically motivated enhancement that adds realism but is not required for bias elimination.

3.5 Toy Model: Consistent with General Mechanism

To verify that the spurious steady state is not an artifact of cardiovascular physiology, we ran the minimal toy model (Section 2.5) under identical dt-sweep conditions. Figure 5 shows the results side by side with the Virtual Vet data.

Table 4: Toy model vs Virtual Vet — identical bias pattern across dt sweep

dt (s)	Toy buggy	Toy fixed	VT pre-fix	VT post-fix	Pattern
0.100	120.0	102.0	111.0	102.3	Bias $\sim 1/\text{dt}$
0.050	140.0	102.0	121.1	102.3	Bias $\sim 1/\text{dt}$
0.020	180.0	102.0	144.7	102.3	Saturated
0.010	180.0	102.0	144.7	102.3	Saturated

The agreement is qualitative but exact in pattern: in both systems, the buggy version produces bias that scales as $\sim 1/\text{dt}$ in the unsaturated regime (bias $\times \text{dt} = 1.90 \pm 0.05$ in the toy model, 1.14 ± 0.04 in Virtual Vet), and both systems plateau at a saturation ceiling when the state variable reaches its bound. The fixed versions in both cases converge to a dt-independent steady state.

The toy model contains no cardiovascular parameters — only a continuous relaxation ($k = 0.25$), a setpoint ($x_0 = 100$), a discrete event magnitude ($K = 0.5$), and a saturation ceiling ($x_{\text{ceiling}} = 180$). The fact that it reproduces the entire bias pattern (unsaturated drift, saturated plateau, dt invariance of the fixed version) is consistent with the hypothesis that the spurious steady state mechanism is intrinsic to the coupling pattern [13] — threshold-gated discrete events emitting dimensionless per-step increments into a continuously integrated state variable — rather than contingent on baroreflex physiology, multi-organ coupling, or any domain-specific detail. A single parameter set is illustrative rather than exhaustive; demonstrating generality across parameter regimes and alternative coupling architectures is a natural direction for further investigation.

4. Discussion

We structure this discussion around three layers of generality. The specific 7-line fix corrects the bug in this codebase. The detection protocol (Section 4.2) generalizes to any modular simulation. The interface fix (Section 4.3) prevents the entire class of failure at the architecture level.

4.1 The Anti-Pattern: dt-Dimensional Ambiguity in Discrete Events

The central finding of this study is that a dt-dimensional mismatch in discrete inter-module events can produce a stable, reproducible steady state that is physiologically incorrect — while passing all standard convergence diagnostics. We term this failure mode spurious steady state: a stable fixed point of the discrete system that does not correspond to any fixed point of the continuous-limit system, arising from a dt-dependent perturbation that vanishes as $dt \rightarrow 0$ but produces a finite offset at any finite dt.

The spurious steady state has three defining characteristics:

1. Stability: The system reaches a steady state with all variables within normal ranges, but at an incorrect operating point.
2. dt-invariance in the saturated regime: Below a critical dt, the pathological steady state becomes independent of dt, giving the false impression that the solution has converged.
3. Diagnostic transparency: Standard

verification methods — dt refinement, steady-state detection, parameter sensitivity analysis — do not trigger alarms because the spurious steady state is robust to all these perturbations.

The mechanism (Figure 1) is a general anti-pattern: Module A emits a value that modifies a continuous state in Module B, but the emission carries no information about whether the value is a rate (normalized to the integration step, unit/s) or an increment (per-step, unit/step). The receiving module cannot distinguish the two because the interface does not declare the temporal dimension. This ambiguity is the anti-pattern's root.

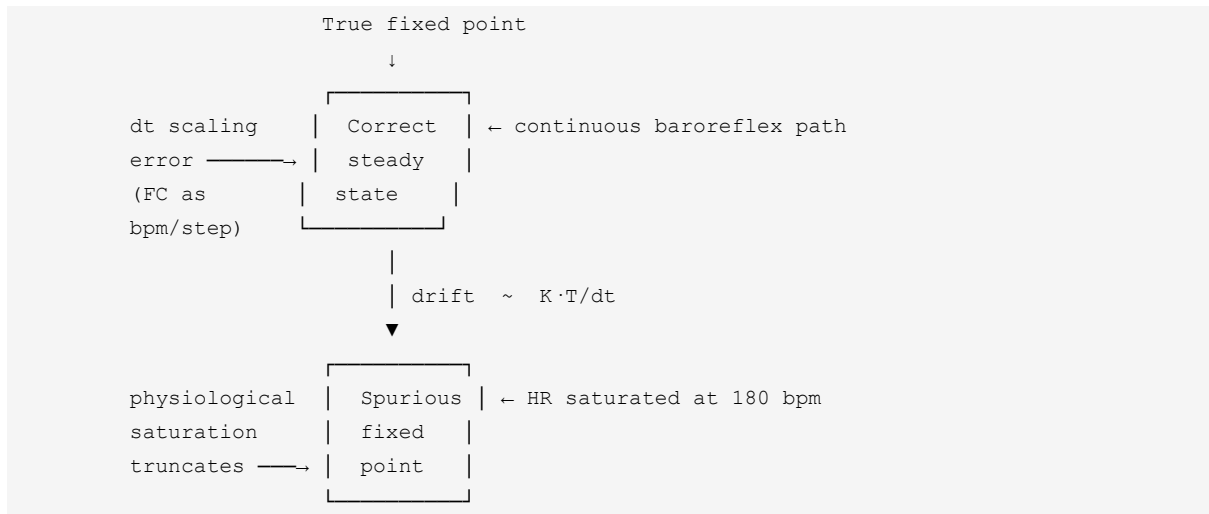


Figure 1: Spurious steady state schematic. The true fixed point is destabilized by the dt-dimensional error; saturation creates a false steady state.

Why the anti-pattern is dangerous — four escape conditions. In our case, the spurious steady state evaded detection through a confluence of four conditions, each independently sufficient to mislead standard diagnostics. Their conjunction explains the

insidiousness:

- Condition 1 — Convergence blind spot from saturation. At $dt \leq 0.02$ s, the HR ceiling (180 bpm) truncated the drift, producing a flat MAP vs dt curve. Standard dt refinement faithfully reports that the discrete solution has stabilized, but it cannot distinguish convergence to the correct fixed point from convergence to a spurious one created by a saturation boundary. - Condition 2 — Module-level unit tests pass. Each organ module was individually correct. The error was not in any module but in the dimensional assumption at the coupling boundary. Cross-module dimensional consistency is not covered by unit testing. - Condition 3 — Face validity of the output. MAP = 144.7 mmHg is elevated but plausible for a canine under stress. The output was a smooth, stable time series — no oscillations, no discontinuities — the hallmarks of numerical instability were entirely absent. - Condition 4 — No standard V&V step covers inter-module dimensional consistency. Code verification checks equation correctness; solution verification estimates discretization error; validation compares against reality. None checks whether a discrete event carries rate dimensions consistent with the receiving module's integration context [11].

The failure of any single condition — no physiological ceiling (removing Condition 1), or MAP reaching 300 mmHg (removing Condition 3) — would have triggered

investigation. That all four held simultaneously created a perfect diagnostic blind spot.

Amplifier: parallel paths. A structural property of the architecture amplified the anti-pattern's hiding power. Two independent mechanisms targeted the same variable (`heart_rate`): the continuous baroreflex path and the discrete FactorCommand path. When one path was removed during testing, the other continued to provide drive, masking the dimensional error. In any modular simulation where multiple modules can write to the same state variable, redundant paths create a testing blind spot.

Generality. The toy model (Section 3.5) proves that this anti-pattern requires no physiological detail — only continuous integration + threshold-gated discrete events + a saturation boundary. The FactorCommand pattern (`target`, `op`, `value`) is domain-neutral, appearing in robotic control, co-simulation, and DEVS-based frameworks [3]. Any modular simulation with these ingredients is susceptible.

4.2 Detection Protocol: Three Levels

The escape analysis above suggests three complementary detection methods, applicable to any modular simulation with discrete-event coupling:

Level 1 — Static dimensional linting (code time). Every discrete event emission point

should be checked at code time: if the value is additive, it must contain a dt factor (rate, unit/s) rather than being a bare increment (unit/step); if multiplicative, it must be exponentiated by dt or expressed as $1 + \text{rate} \times \text{dt}$. We have released a static lint tool (`check_fc_dimensions.py`) that automates this check for the FactorCommand pattern, and it successfully detected a second latent issue in the immune module after the primary fix.

Level 2 — Runtime dt-sweep audit (integration time). A wrapper procedure: run the simulation at dt, 2dt, 4dt, and 8dt, and compute $|\text{SS}(\text{dt}) - \text{SS}(8\text{dt})|$. If the relative variation exceeds 1% for any key output, flag for dimensional review. In our data, the buggy engine showed MAP range 33.7 mmHg (>30% variation across dt), while the fixed engine showed MAP range 2.21 mmHg (~2%). This test directly targets the signature of dimensional inconsistency — systematic drift with step size — with minimal implementation effort.

Level 3 — Interface contract metadata (design time). The most fundamental level: make the temporal dimension of every discrete event a first-class part of the interface (see Section 4.3 for detailed discussion). Levels 1 and 2 detect existing errors; Level 3 prevents them from being introduced.

Negative result that is itself diagnostic. An important finding is what does NOT help:

ordering swap tests (heart→neuro vs neuro→heart) showed $\Delta < 0.034$ mmHg, which might suggest the architecture is insensitive to coupling effects. It is not — the swap test is simply not diagnostic for dimensional errors. Ordering sensitivity and dimensional inconsistency are distinct failure modes; the absence of one does not rule out the other.

4.3 The Interface Fix: Preventing the Class

The 7-line code change described in Section 3.4 (multiplying FC deltas by dt) fixes this specific instance, but it is a case-specific remedy, not a class-level prevention. The bug was possible because the FactorCommand interface carried no information about the temporal interpretation of the value. The fix that prevents the entire class of failure is to make temporal type a first-class component of the inter-module interface contract.

Proposal: temporal type annotation on discrete events. Every discrete event type should declare whether its value is a rate (normalized to the integration step) or an increment (per-step):

```
@dataclass
class DiscreteEvent:
    target: str
    op: Literal["add", "multiply"]
    value: float
    # Temporal metadata (proposed)
    temporal_type: Literal["rate", "increment"] # per-second vs per-step
    unit: str # bpm, mmHg, L/min ...
```

With this declaration, the simulation engine's dispatch layer enforces the constraint at

runtime: a "rate" event that has not been scaled by dt before reaching the application point is flagged. At design time, the annotation makes the temporal dimension explicit in code review. At code time, a lint rule (Level 1) verifies consistency between the declared `temporal_type` and the actual expression producing value.

Why existing interface standards are insufficient. The Functional Mock-up Interface (FMI) [14] — the dominant standard for modular simulation — annotates every variable with physical units via the `BaseUnit` element (kg, m, s, etc.). When two variables are connected, FMI checks that their exponent vectors match, catching mismatches such as connecting a force ($\text{kg}\cdot\text{m}/\text{s}^2$) to an angle (rad). This is valuable, but it cannot distinguish a rate (bpm/s) from an increment (bpm/step) because both share the physical dimension $1/\text{time}$ — the distinction is not in the SI unit system but in whether the value is normalized by the integration step size. Nutaro et al. [13] identified this same limitation in their analysis of time representation in hybrid co-simulation and proposed FMI-HC extensions using integer time and absent signals for discrete event handling — a complementary direction that addresses temporal resolution but, like the base FMI standard, does not capture the rate-versus-increment dimension. Modelica's unit checking [15], the most advanced dimensional inference in simulation software, similarly operates on physical dimensions only.

The `temporal_type` annotation fills this gap. It addresses a dimension — rate vs increment — that the SI system does not express and that existing interface standards do not capture. This is a natural extension of the design-by-contract principle [16] applied to simulation coupling: the interface should specify not just what physical quantity is exchanged, but how its value is temporally qualified relative to the integration process.

4.4 Relationship to Prior Work

Our results complement and extend previous findings on numerical bias in physiological simulation. Tłalka et al. [5] reported that explicit Euler methods are "numerically unstable" for baroreflex models; we identify a specific, preventable mechanism for this instability — not a fundamental property of explicit Euler, but a dimensional error in discrete inter-module events.

The spurious steady state phenomenon is distinct from the splitting error analyzed by Kim et al. (2011) in poromechanics. In Kim's analysis, the error arises from the mathematical structure of the operator split and persists as $dt \rightarrow 0$ — an intrinsic property of the sequential coupling scheme. In our case, the error would vanish if the discrete event were correctly normalized; it is a dimensional inconsistency in the event emission protocol, not a fundamental limitation of sequential integration. The distinction is important: Kim's error requires architectural changes (e.g., monolithic coupling) to

eliminate, whereas ours requires only an interface-level temporal annotation.

Co-simulation coupling error research [17, 18] addresses errors from discrete-time information exchange — input extrapolation, signal reconstruction, energy artifacts at macro-step boundaries. These are temporal discretization artifacts, distinct from the temporal dimensional ambiguity we identify. The former concerns when information is exchanged; the latter concerns what temporal interpretation the exchanged value carries.

Nutaro et al. [13] proposed a split system approach for managing time in hybrid simulations combining continuous and discrete-event components, classifying hybrid systems into essentially continuous, essentially discrete, and truly hybrid — where applying standard packages leads to inaccuracies. Their constructive methodology uses a priori knowledge of model structure to decompose and re-articulate hybrid systems for existing simulation tools. Our work complements theirs from the diagnostic direction: where Nutaro provides a design methodology for getting hybrid time management right, we document a specific failure mode that arises when it goes wrong and propose the temporal type metadata that would make such failures detectable at the interface level.

4.5 Limitations

This study has several limitations. First, the Virtual Vet results are from a single

simulation platform. The toy model (Section 3.5) confirms the spurious steady state pattern is intrinsic to the coupling design rather than platform-specific, but whether it manifests in other production-grade engines (HumMod, SAPHIR, CellML-based frameworks) with their specific event architectures remains to be systematically investigated. Second, the clinical validation is limited to comparison with a single published canine hypoxia study; a broader validation against multiple physiological scenarios is warranted. Third, the hemorrhage-induced ordering sensitivity ($\Delta = 2.84$ mmHg during transient) is a genuine property of the sequential Euler architecture that persists after the dt-scaling fix; it represents a separate, unresolved issue outside the scope of the present work. Fourth, the Y/Z isolation confound (Section 3.4 note) limits the precision of the B_contrib attribution, though it does not affect the primary finding that Operation A alone eliminates the bias.

All simulations use IEEE 754 double-precision arithmetic. The subprocess-isolated cross-run agreement ($\Delta = 0.034$ mmHg, Section 3.2) confirms that floating-point accumulation order does not affect the reported results at the precision level relevant to this study.

5. Conclusion

We documented the discovery, diagnosis, and correction of a dimensional analysis error

in one FactorCommand channel of the Virtual Vet canine cardiovascular simulation: a heart rate modifier implemented as a fixed per-step increment (bpm/step) rather than a time-normalized rate (bpm/s), producing a steady-state MAP bias of +44.7 mmHg that standard convergence diagnostics could not detect. The bias scaled as $\sim 1/\text{dt}$ before saturating at the HR physiological ceiling of 180 bpm — a spurious steady state.

This study identifies a general anti-pattern: dimensional ambiguity in discrete-event interfaces — the coupling interface does not declare whether a transmitted value is a rate (per-second) or an increment (per-step). The anti-pattern evades detection because it combines four escape conditions (saturation blind spot, passing module tests, plausible output, and cross-module dimensions outside standard V&V scope) and is amplified by parallel redundant paths. We propose a three-level detection protocol: static dimensional linting (code time), dt-sweep audit (integration time), and interface contract metadata (design time). The fundamental interface fix is not the 7-line dt-scaling correction — which resolves only this instance — but the addition of temporal type metadata to the discrete-event interface contract, a dimension that existing standards (FMI, Modelica) do not cover.

Whether this failure pattern recurs in other simulation platforms (HumMod, SAPHIR, CellML-based frameworks) is an open question warranting systematic investigation.

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Figures

Figure 1: Spurious steady state schematic — drift and saturation mechanism ![Figure 1](fig1_pseudo_convergence_schematic.png)

Figure 2: MAP bias vs dt (log-log) — pre-fix, three regimes ![Figure 2](fig2_map_bias_vs_dt.png)

Figure 3: Code diff — the 7-line fix ![Figure 3](fig3_code_diff.png)

Figure 4: Before/after comparison — X/Y/Z MAP range bar chart across DC conditions ![Figure 4](fig4_before_after_comparison.png)

Figure 5: Toy model vs Virtual Vet — identical bias pattern. (a) Minimal 2-variable ODE with same coupling pattern shows bias $\sim 1/dt$ in the unsaturated regime and saturation plateau; (b) Virtual Vet results overlaid for comparison ![Figure 5](fig5_toy_model_comparison.png)

Data and Code Availability

The Virtual Vet simulation engine, experiment scripts (exp6–exp9), and experimental data (JSON files) are available at <<https://github.com/ningjie333/virtual-vet>>. All experiments can be reproduced by running the Python scripts in the experiments/ directory with Python 3.13+ and the dependencies listed in pyproject.toml. The static lint tool `check_fc_dimensions.py` is included in the repository under tools/.